

Eigenvalues, PCA, and Dimensionality Reduction

Covariance geometry, components, dimensionality reduction, and leakage-safe PCA workflows.

Library of the Quant

Educational reference manual

Manual Profile

LIBRARY ID	LOTQ-0018
CATEGORY	Linear Algebra
DIFFICULTY	Intermediate
FORMAT	Single-column field-guide manual
USE	Study, lookup, and research-system reference

Educational Use Only

This manual is educational material for quant research study. It is not financial advice, not a trading recommendation, and not a guarantee of correctness or live trading usefulness.

Table of contents

INTERMEDIATE

Title	1
Table of contents I	2
Table of contents II	3
How to use this manual	4
Notation and shape discipline	5
Leakage-safe PCA workflow	6
Vectors as market states	7
Matrices as transformations	8
Basis and coordinates	9
Dot products and projection	10
Orthogonality	11
Matrix rank	12
Covariance geometry	13
Eigenvalue equation	14
Eigenvalues	15
Eigenvectors	16
Symmetric matrices	17
Positive semidefinite covariance	18
Spectral decomposition	19
Trace and total variance	20
Condition number	21
Singular value decomposition	22
Centering	23
Standardization	24
Covariance PCA vs correlation PCA	25
Principal component analysis	26

Table of contents

INTERMEDIATE

Component scores	27
Loadings	28
Explained variance ratio	29
Scree plots	30
Choosing component count	31
Reconstruction	32
Reconstruction error	33
Low-rank approximation	34
Whitening	35
PCA as denoising	36
Rolling and expanding PCA	37
Component instability	38
Sign ambiguity and alignment	39
Outliers and robust PCA	40
Missing data and asynchronous markets	41
PCA and factor models	42
Leakage-safe research pipeline	43
Production monitoring	44
Covariance geometry visual lab	45
Scree plot decision lab	46
Regime stability heatmap	47
Formula review	48
Diagnostic review	49
Implementation checklist	50
Final active recall index	51

How to use this manual

INTERMEDIATE

1. SCAN FOR RECALL

Start with the meaning and memory-jogger cards. If you can explain the concept in your own words, move directly to the formula and failure-mode panels.

2. TRACE THE GEOMETRY

For every equation, name the dimensions and units. Ask what rotates, what scales, what is projected, and what information is discarded.

3. CONNECT TO RESEARCH

Translate the concept into a market-data example, then identify where it belongs in the research pipeline and what artifacts must be stored.

4. VALIDATE TEMPORALLY

Any learned mean, scale, covariance, basis, component count, or imputation rule must be fit only on the training window.

5. REVIEW ACTIVELY

Answer the quick-recall prompt before rereading. Recreate worked calculations and implement small synthetic tests before touching live data.

READING CONVENTION

X is observations by features ($T \times p$). Centered data is X_c . V holds feature directions. Z contains projected scores. k is the retained dimension.

DECISION STANDARD

PCA is useful only when the compressed representation improves a defined outcome: stability, speed, reconstruction, risk estimation, or validated model performance.

Notation and shape discipline

INTERMEDIATE

SYMBOL	SHAPE	MEANING
X	$T \times p$	Raw observation-by-feature matrix
X_c	$T \times p$	Centered training or transformed data
mu	$p \times 1$	Training feature means
Sigma	$p \times p$	Covariance matrix
V_k	$p \times k$	Retained orthonormal component directions
Lambda_k	$k \times k$	Retained eigenvalues on the diagonal
Z	$T \times k$	Component scores
X_hat	$T \times p$	Reconstructed approximation
w	$p \times 1$	Portfolio or linear-combination weights
lambda_i	scalar	Variance along component i

SHAPE AUDIT BEFORE MULTIPLICATION

For $Z = X_c V_k$: $(T \times p)(p \times k) = T \times k$. For reconstruction $X_{hat} = Z V_k^T + \mu$: $(T \times k)(k \times p) + \text{broadcast}(p) = T \times p$. Write this audit beside production code and tests; it prevents silent axis swaps.

UNIT AUDIT

Covariance eigenvalues carry squared feature units. Correlation-PCA eigenvalues are unitless. Scores inherit the scaled input convention. Any comparison across experiments must first confirm identical features, ordering, scaling, and estimator windows.

Leakage-safe PCA research workflow

INTERMEDIATE

1

Define decision

Target, horizon, universe, latency

2

Build panel

Calendars, missingness, bad ticks

3

Split time

Train, validation, test; purge if needed

4

Fit transforms

Imputer, center, scale, PCA on train

5

Select k

Nested validation and stability tests

6

Evaluate

Reconstruction plus downstream outcomes

7

Register

Means, scales, basis, signs, schema

8

Monitor

Scores, residuals, angles, performance

HARD GATE

No fitted statistic may see future observations. A pipeline that fits scaling or PCA on the full dataset is invalid even if the predictive model itself is trained fold by fold.

Vectors as market states

INTERMEDIATE

ONE-SENTENCE MEANING

A vector is an ordered snapshot of measurements, such as returns across assets at one timestamp.

MEMORY JOGGER

Treat each coordinate as a named axis, not merely a number. Changing the asset order changes the meaning.

WHY IT MATTERS IN QUANT RESEARCH

Cross-sectional returns, factor exposures, and model features are all vectors that can be compared, projected, and transformed.

FORMULA INTUITION

$x = [x_1, \dots, x_p]^T$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

Five daily returns form one point in a five-dimensional feature space; nearby points represent similar market states.

PYTHON / SYSTEM IMPLEMENTATION

Store a stable feature order with the array and assert its schema before every matrix operation.

CONFUSIONS TO AVOID

Do not mix units or silently reorder columns. Standardization may be necessary before distance or PCA.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to basis, norm, dot product, feature matrix.

QUICK RECALL

Explain vectors as market states without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Matrices as transformations

INTERMEDIATE

ONE-SENTENCE MEANING

A matrix can store data or act as a rule that maps one vector into another.

MEMORY JOGGER

Rows and columns have roles. Write those roles down before multiplying anything.

WHY IT MATTERS IN QUANT RESEARCH

Matrices represent return panels, covariance, factor loadings, linear models, and portfolio constraints.

FORMULA INTUITION

$y = A x$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A 3×5 loading matrix converts five asset returns into three factor scores.

PYTHON / SYSTEM IMPLEMENTATION

Use explicit shapes in code comments and tests: X is $T \times p$, weights are $p \times 1$, scores are $T \times k$.

CONFUSIONS TO AVOID

A dimensionally valid multiplication can still be conceptually backward; shape checks are necessary but not sufficient.

WHERE THIS FITS

Research path: clean data $\rightarrow$ fit transformation $\rightarrow$ validate out of sample $\rightarrow$ store artifacts $\rightarrow$ monitor drift. This concept connects directly to linear map, transpose, inverse, rank.

QUICK RECALL

Explain matrices as transformations without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Basis and coordinates

INTERMEDIATE

ONE-SENTENCE MEANING

A basis is a set of independent directions used to describe every vector in a space.

MEMORY JOGGER

The point stays the same while its coordinates change when you rotate the axes.

WHY IT MATTERS IN QUANT RESEARCH

PCA replaces the original feature basis with orthogonal directions ordered by variance.

FORMULA INTUITION

$x = B a$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

An asset-return vector can be described in asset coordinates or in principal-component coordinates.

PYTHON / SYSTEM IMPLEMENTATION

Persist the fitted basis matrix with model metadata; new data must use the same basis and preprocessing.

CONFUSIONS TO AVOID

PCA coordinates are not original features and may change sign without changing the subspace.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to span, linear independence, orthogonality.

QUICK RECALL

Explain basis and coordinates without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Dot products and projection

INTERMEDIATE

ONE-SENTENCE MEANING

The dot product measures directional alignment and produces projection coordinates.

MEMORY JOGGER

Projection asks: how much of x points along direction v ?

WHY IT MATTERS IN QUANT RESEARCH

PCA scores are projections of centered observations onto eigenvector directions.

FORMULA INTUITION

$\text{score} = x^T v$; $\text{proj}_v(x) = (x^T v / v^T v) v$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

If a normalized market-mode vector has score 2.1, the observation has strong exposure in that direction.

PYTHON / SYSTEM IMPLEMENTATION

Use matrix multiplication `X_centered @ components.T`; verify component normalization.

CONFUSIONS TO AVOID

Raw dot products depend on scale. Cosine similarity removes magnitude, but PCA projection intentionally retains it.

WHERE THIS FITS

Research path: clean data $\rightarrow$ fit transformation $\rightarrow$ validate out of sample $\rightarrow$ store artifacts $\rightarrow$ monitor drift. This concept connects directly to orthogonality, cosine similarity, residual.

QUICK RECALL

Explain dot products and projection without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Orthogonality

INTERMEDIATE

ONE-SENTENCE MEANING

Two vectors are orthogonal when their dot product is zero.

MEMORY JOGGER

Orthogonal directions carry no linear overlap in Euclidean geometry.

WHY IT MATTERS IN QUANT RESEARCH

PCA components are orthogonal, which simplifies attribution and reconstruction.

FORMULA INTUITION

$u^T v = 0$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A market component and a sector-spread component can be constructed to have zero sample covariance in score space.

PYTHON / SYSTEM IMPLEMENTATION

Numerically check $V @ V.T$ against the identity matrix within a tolerance.

CONFUSIONS TO AVOID

Orthogonal does not mean statistically independent except under stronger distributional assumptions.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to uncorrelatedness, independence, Gram-Schmidt.

QUICK RECALL

Explain orthogonality without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Matrix rank

INTERMEDIATE

ONE-SENTENCE MEANING

Rank counts the number of independent directions present in a matrix.

MEMORY JOGGER

Rank is the effective number of axes the data truly occupies.

WHY IT MATTERS IN QUANT RESEARCH

Collinear indicators, duplicate assets, and short samples create rank deficiency and unstable estimates.

FORMULA INTUITION

$\text{rank}(X) \leq \min(T, p)$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

Ten indicators built from the same price series may span only three or four meaningful directions.

PYTHON / SYSTEM IMPLEMENTATION

Inspect singular values and use numerical rank with an explicit tolerance rather than exact equality.

CONFUSIONS TO AVOID

A tiny singular value is practically zero even if floating-point arithmetic reports full rank.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to null space, condition number, SVD.

QUICK RECALL

Explain matrix rank without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Covariance geometry

INTERMEDIATE

ONE-SENTENCE MEANING

A covariance matrix describes the scale and orientation of a cloud of centered observations.

MEMORY JOGGER

Its eigenvectors are ellipse axes; eigenvalues are variance along those axes.

WHY IT MATTERS IN QUANT RESEARCH

Portfolio risk, factor extraction, and PCA all depend on covariance geometry.

FORMULA INTUITION

$\Sigma = (1/(T-1)) X_c^T X_c$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A highly correlated equity basket forms an elongated cloud with one dominant market direction.

PYTHON / SYSTEM IMPLEMENTATION

Estimate covariance on the training window only; consider shrinkage when p is large relative to T .

CONFUSIONS TO AVOID

Covariance is unit-sensitive and noisy. Correlation removes units but changes the optimization problem.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to correlation, ellipsoid, shrinkage.

QUICK RECALL

Explain covariance geometry without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Eigenvalue equation

INTERMEDIATE

ONE-SENTENCE MEANING

An eigenvector keeps its direction under a linear transformation; its eigenvalue is the scale factor.

MEMORY JOGGER

The matrix stretches this special direction without turning it.

WHY IT MATTERS IN QUANT RESEARCH

For covariance matrices, eigenvectors define variance directions and eigenvalues quantify their variance.

FORMULA INTUITION

$A v = \lambda v$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

If $\Sigma v = 0.014 v$, a portfolio aligned with v has variance 0.014 when v has unit length.

PYTHON / SYSTEM IMPLEMENTATION

Use `eigh` for symmetric covariance matrices; it is more stable and returns real eigenpairs.

CONFUSIONS TO AVOID

Eigenvectors are defined up to sign, and repeated eigenvalues do not identify a unique basis.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to characteristic polynomial, eigenspace, spectrum.

QUICK RECALL

Explain eigenvalue equation without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Eigenvalues

INTERMEDIATE

ONE-SENTENCE MEANING

Eigenvalues summarize how strongly a matrix acts along its eigenvector directions.

MEMORY JOGGER

Large means strong stretch; near zero means little independent information.

WHY IT MATTERS IN QUANT RESEARCH

In PCA, eigenvalues rank components by explained sample variance.

FORMULA INTUITION

$\det(A - \lambda I) = 0$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

Eigenvalues [5.2, 1.1, 0.2] indicate one dominant direction and one weak direction.

PYTHON / SYSTEM IMPLEMENTATION

Sort descending for PCA and clip only tiny negative values caused by numerical noise, not genuine model problems.

CONFUSIONS TO AVOID

Eigenvalue magnitude depends on feature scaling and the covariance estimator.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to trace, determinant, spectral decomposition.

QUICK RECALL

Explain eigenvalues without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Eigenvectors

INTERMEDIATE

ONE-SENTENCE MEANING

Eigenvectors are invariant directions associated with eigenvalues.

MEMORY JOGGER

They are directions, not observations and not automatically economic factors.

WHY IT MATTERS IN QUANT RESEARCH

Loadings reveal which standardized features move together along each component.

FORMULA INTUITION

$V^T V = I$ for symmetric matrices. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

An eigenvector with similar positive weights across equities often resembles a broad market mode.

PYTHON / SYSTEM IMPLEMENTATION

Align signs to a reference convention for readable reports, but never treat sign flips as model drift.

CONFUSIONS TO AVOID

Coefficient magnitude reflects direction composition; it is not a regression coefficient or causal effect.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to loadings, scores, rotation.

QUICK RECALL

Explain eigenvectors without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Symmetric matrices

INTERMEDIATE

ONE-SENTENCE MEANING

A real symmetric matrix has real eigenvalues and an orthonormal eigenbasis.

MEMORY JOGGER

Symmetry gives clean geometry and stable specialized algorithms.

WHY IT MATTERS IN QUANT RESEARCH

Sample covariance and correlation matrices should be symmetric, making PCA especially tractable.

FORMULA INTUITION

$A = A^T$; $A = V \Lambda V^T$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A covariance matrix can be reconstructed exactly from its eigenvectors and eigenvalues, apart from rounding.

PYTHON / SYSTEM IMPLEMENTATION

Symmetrize only tiny floating-point discrepancies; investigate meaningful asymmetry upstream.

CONFUSIONS TO AVOID

Not every symmetric-looking estimate is positive semidefinite after missing-data pairwise calculations.

WHERE THIS FITS

Research path: clean data $\rightarrow$ fit transformation $\rightarrow$ validate out of sample $\rightarrow$ store artifacts $\rightarrow$ monitor drift. This concept connects directly to positive semidefinite, spectral theorem.

QUICK RECALL

Explain symmetric matrices without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Positive semidefinite covariance

INTERMEDIATE

ONE-SENTENCE MEANING

A covariance matrix is positive semidefinite because every portfolio variance must be nonnegative.

MEMORY JOGGER

No direction can have negative squared spread.

WHY IT MATTERS IN QUANT RESEARCH

PSD structure is required for coherent risk, optimization, and real-valued PCA variance.

FORMULA INTUITION

$w^T \Sigma w \geq 0$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A negative eigenvalue of material size signals a broken covariance construction or inconsistent data treatment.

PYTHON / SYSTEM IMPLEMENTATION

Check minimum eigenvalue and condition number before optimization; use principled PSD repair if required.

CONFUSIONS TO AVOID

Blind eigenvalue clipping can hide data-quality errors and distort risk.

WHERE THIS FITS

Research path: clean data $\rightarrow$ fit transformation $\rightarrow$ validate out of sample $\rightarrow$ store artifacts $\rightarrow$ monitor drift. This concept connects directly to Cholesky, nearest PSD, regularization.

QUICK RECALL

Explain positive semidefinite covariance without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Spectral decomposition

INTERMEDIATE

ONE-SENTENCE MEANING

Spectral decomposition rebuilds a symmetric matrix from orthogonal eigenvectors and eigenvalues.

MEMORY JOGGER

Rotate, scale each axis, then rotate back.

WHY IT MATTERS IN QUANT RESEARCH

It separates covariance into independent variance modes for analysis, compression, and simulation.

FORMULA INTUITION

$\Sigma = \sum_i \lambda_i v_i v_i^T$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

Keeping the top two terms gives the best rank-two covariance approximation under common matrix norms.

PYTHON / SYSTEM IMPLEMENTATION

Reconstruction tests catch transposition, sorting, and scaling mistakes.

CONFUSIONS TO AVOID

Truncation removes low-variance structure, which may still contain useful alpha or hedge information.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to low-rank approximation, matrix functions.

QUICK RECALL

Explain spectral decomposition without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Trace and total variance

INTERMEDIATE

ONE-SENTENCE MEANING

The trace of a covariance matrix equals total feature variance and the sum of eigenvalues.

MEMORY JOGGER

Diagonal variance and spectral variance are the same total seen in two coordinate systems.

WHY IT MATTERS IN QUANT RESEARCH

Explained-variance ratios divide each eigenvalue by this total.

FORMULA INTUITION

$\text{tr}(\Sigma) = \sum_i \lambda_i$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

Eigenvalues 4, 2, 1 explain 4/7, 2/7, and 1/7 of total variance.

PYTHON / SYSTEM IMPLEMENTATION

Assert that eigenvalue sums match the covariance trace within numerical tolerance.

CONFUSIONS TO AVOID

Total variance is meaningful only under the chosen scaling and feature set.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to explained variance, scree plot.

QUICK RECALL

Explain trace and total variance without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Condition number

INTERMEDIATE

ONE-SENTENCE MEANING

The condition number measures sensitivity to small perturbations by comparing largest and smallest singular scales.

MEMORY JOGGER

A long, thin geometry amplifies estimation noise.

WHY IT MATTERS IN QUANT RESEARCH

Ill-conditioned covariance destabilizes inverse-risk weights, regressions, and tiny PCA directions.

FORMULA INTUITION

$\kappa(A) = \sigma_{\max} / \sigma_{\min}$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A condition number near 10^8 warns that small data changes can radically alter inverse-based results.

PYTHON / SYSTEM IMPLEMENTATION

Log condition numbers through time and set review thresholds tied to precision and use case.

CONFUSIONS TO AVOID

There is no universal safe cutoff; consequences depend on downstream operations.

WHERE THIS FITS

Research path: clean data $\rightarrow$ fit transformation $\rightarrow$ validate out of sample $\rightarrow$ store artifacts $\rightarrow$ monitor drift. This concept connects directly to multicollinearity, regularization, pseudoinverse.

QUICK RECALL

Explain condition number without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Singular value decomposition

INTERMEDIATE

ONE-SENTENCE MEANING

SVD factors any matrix into left directions, nonnegative scales, and right directions.

MEMORY JOGGER

SVD is PCA's direct route from the data matrix without explicitly forming covariance.

WHY IT MATTERS IN QUANT RESEARCH

It is numerically stable and works when features exceed observations.

FORMULA INTUITION

$X_c = U S V^T$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

For covariance PCA, eigenvalues equal $s_i^2/(T-1)$, and rows of V^T are component directions.

PYTHON / SYSTEM IMPLEMENTATION

Prefer economy SVD for large dense matrices; randomized SVD is useful when only a few components are needed.

CONFUSIONS TO AVOID

Do not confuse left singular vectors (time scores after scaling) with right singular vectors (feature directions).

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to eigendecomposition, truncated SVD.

QUICK RECALL

Explain singular value decomposition without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Centering

INTERMEDIATE

ONE-SENTENCE MEANING

Centering subtracts the training mean so PCA describes variation around the sample center.

MEMORY JOGGER

Move the cloud to the origin before finding its axes.

WHY IT MATTERS IN QUANT RESEARCH

Without centering, the first component may mainly point toward the mean rather than explain fluctuations.

FORMULA INTUITION

$X_c = X - 1 \mu^T$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

Subtract each feature's training-window mean, then apply the same stored means to validation data.

PYTHON / SYSTEM IMPLEMENTATION

Fit the centering transform inside each training fold to prevent look-ahead leakage.

CONFUSIONS TO AVOID

Re-centering every live observation with future data changes history and invalidates backtests.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to demeaning, intercept, preprocessing.

QUICK RECALL

Explain centering without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Standardization

INTERMEDIATE

ONE-SENTENCE MEANING

Standardization divides centered features by a scale estimate, usually sample standard deviation.

MEMORY JOGGER

Give features comparable units before asking variance to choose importance.

WHY IT MATTERS IN QUANT RESEARCH

It prevents high-volatility or large-unit variables from dominating correlation-based PCA.

FORMULA INTUITION

$z_j = (x_j - \mu_j) / s_j$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

Volatility at 0.02 and volume at millions become comparable after standardization.

PYTHON / SYSTEM IMPLEMENTATION

Store means, scales, estimator choices, and zero-variance handling with the fitted pipeline.

CONFUSIONS TO AVOID

Standardization can erase economically meaningful scale; choose covariance or correlation PCA deliberately.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to robust scaling, winsorization, normalization.

QUICK RECALL

Explain standardization without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Covariance PCA vs correlation PCA

INTERMEDIATE

ONE-SENTENCE MEANING

Covariance PCA preserves original scale; correlation PCA standardizes each feature first.

MEMORY JOGGER

Choose whether scale is signal or nuisance.

WHY IT MATTERS IN QUANT RESEARCH

Return series with comparable units may use covariance; mixed indicators usually need standardization.

FORMULA INTUITION

$R = D^{(-1/2)} \Sigma D^{(-1/2)}$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A volatile asset dominates covariance PCA but receives equal marginal variance under correlation PCA.

PYTHON / SYSTEM IMPLEMENTATION

Make the choice explicit in experiment metadata and compare component stability under both.

CONFUSIONS TO AVOID

Correlation PCA is not automatically superior; equalizing noise can elevate weak features.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to feature scaling, whitening.

QUICK RECALL

Explain covariance pca vs correlation pca without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Principal component analysis

INTERMEDIATE

ONE-SENTENCE MEANING

PCA rotates centered data into orthogonal directions ordered by explained variance.

MEMORY JOGGER

Find the widest axis, remove its projection, then repeat in the remaining space.

WHY IT MATTERS IN QUANT RESEARCH

PCA compresses correlated features, reveals common modes, and stabilizes some downstream models.

FORMULA INTUITION

$v_1 = \operatorname{argmax}_{\{||v||=1\}} \operatorname{Var}(X_c v)$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A universe of correlated equity returns may be summarized by market, sector, and style-like components.

PYTHON / SYSTEM IMPLEMENTATION

Use a pipeline: impute, scale, fit PCA on training data, transform validation, then model.

CONFUSIONS TO AVOID

PCA maximizes variance, not predictability, causality, interpretability, or trading performance.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to scores, loadings, explained variance.

QUICK RECALL

Explain principal component analysis without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Component scores

INTERMEDIATE

ONE-SENTENCE MEANING

A component score is an observation's coordinate after projection onto a principal direction.

MEMORY JOGGER

Loadings define the axis; scores tell where each observation lies on it.

WHY IT MATTERS IN QUANT RESEARCH

Scores become compressed model features, regime indicators, or reconstructed factor returns.

FORMULA INTUITION

$Z = X_c V_k$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A large negative first score can indicate a broad risk-off day if the sign convention is fixed.

PYTHON / SYSTEM IMPLEMENTATION

Transform with frozen preprocessing and component matrices; monitor score distribution drift.

CONFUSIONS TO AVOID

Scores depend on input scaling and eigenvector sign; only relative geometry is invariant.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to factor returns, projection, latent variables.

QUICK RECALL

Explain component scores without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Loadings

INTERMEDIATE

ONE-SENTENCE MEANING

Loadings describe how original variables contribute to a component direction.

MEMORY JOGGER

Read across a component to see its feature recipe.

WHY IT MATTERS IN QUANT RESEARCH

They help label common modes and diagnose concentration in a few features.

FORMULA INTUITION

$\text{loading}_{jk} = v_{jk}$ (or $v_{jk} \sqrt{\lambda_k}$, by convention). Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

Nearly equal positive equity weights suggest a broad market direction; opposing signs may describe a spread.

PYTHON / SYSTEM IMPLEMENTATION

Document the loading convention because libraries and disciplines use the word differently.

CONFUSIONS TO AVOID

Large absolute loadings do not prove economic importance or causal influence.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to eigenvectors, correlations with scores.

QUICK RECALL

Explain loadings without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Explained variance ratio

INTERMEDIATE

ONE-SENTENCE MEANING

Explained variance ratio is the fraction of total sample variance assigned to a component.

MEMORY JOGGER

It measures compression coverage, not forecasting value.

WHY IT MATTERS IN QUANT RESEARCH

Cumulative ratios help choose a candidate dimension and communicate information retained.

FORMULA INTUITION

$EVR_i = \lambda_i / \sum_j \lambda_j$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

Ratios 0.55, 0.22, 0.10 retain 87% with three components.

PYTHON / SYSTEM IMPLEMENTATION

Compute ratios on training data and report validation reconstruction error separately.

CONFUSIONS TO AVOID

A high ratio may reflect one volatile nuisance feature or a market mode unrelated to alpha.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to scree plot, cumulative variance.

QUICK RECALL

Explain explained variance ratio without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Scree plots

INTERMEDIATE

ONE-SENTENCE MEANING

A scree plot graphs eigenvalues or explained-variance ratios against component rank.

MEMORY JOGGER

Look for the elbow where steep information gains flatten.

WHY IT MATTERS IN QUANT RESEARCH

It supports dimension choice and exposes dominant common structure.

FORMULA INTUITION

plot $i \rightarrow \lambda_i$ or cumulative EVR. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A sharp drop after component three suggests a compact three-mode description, subject to validation.

PYTHON / SYSTEM IMPLEMENTATION

Include uncertainty or rolling-window overlays rather than relying on one static curve.

CONFUSIONS TO AVOID

Elbows are subjective; a smooth spectrum may have no defensible visual cutoff.

WHERE THIS FITS

Research path: clean data $\rightarrow$ fit transformation $\rightarrow$ validate out of sample $\rightarrow$ store artifacts $\rightarrow$ monitor drift. This concept connects directly to parallel analysis, cross-validation.

QUICK RECALL

Explain scree plots without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Choosing component count

INTERMEDIATE

ONE-SENTENCE MEANING

Component count balances compression, stability, reconstruction, and downstream usefulness.

MEMORY JOGGER

Keep enough structure for the task, not enough variance for a slogan.

WHY IT MATTERS IN QUANT RESEARCH

The best k for denoising returns may differ from the best k for forecasting or risk modeling.

FORMULA INTUITION

$k^* = \operatorname{argmin} \text{validation loss}(k) \text{ subject to complexity.}$ Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

Compare k from 2 to 20 using walk-forward model loss, turnover, and loading stability.

PYTHON / SYSTEM IMPLEMENTATION

Tune k inside nested time-series validation and log the selection rule before final testing.

CONFUSIONS TO AVOID

Selecting k on the test period leaks information and overstates robustness.

WHERE THIS FITS

Research path: clean data $\rightarrow$ fit transformation $\rightarrow$ validate out of sample $\rightarrow$ store artifacts $\rightarrow$ monitor drift. This concept connects directly to elbow rule, information criteria, nested CV.

QUICK RECALL

Explain choosing component count without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Reconstruction

INTERMEDIATE

ONE-SENTENCE MEANING

Reconstruction maps reduced scores back into the original feature space.

MEMORY JOGGER

Compress, then expand using only retained axes.

WHY IT MATTERS IN QUANT RESEARCH

It quantifies information loss and supports denoising, anomaly detection, and diagnostics.

FORMULA INTUITION

$\hat{X} = Z V_k^T + \mu$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A large residual on one asset can flag an idiosyncratic move not explained by common components.

PYTHON / SYSTEM IMPLEMENTATION

Measure reconstruction error by feature and regime, not only as one global average.

CONFUSIONS TO AVOID

Low reconstruction error does not guarantee preserved predictive signal.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to residual, decoder, low-rank model.

QUICK RECALL

Explain reconstruction without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Reconstruction error

INTERMEDIATE

ONE-SENTENCE MEANING

Reconstruction error measures the distance between original and low-rank reconstructed data.

MEMORY JOGGER

What retained components cannot explain becomes the residual.

WHY IT MATTERS IN QUANT RESEARCH

It can identify anomalies, determine compression adequacy, and compare preprocessing choices.

FORMULA INTUITION

$MSE = ||X - \hat{X}||_F^2 / (T \cdot p)$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A sudden spike in residual error may indicate a new regime, bad data, or a novel idiosyncratic event.

PYTHON / SYSTEM IMPLEMENTATION

Track normalized error and per-feature contributions with train-derived thresholds.

CONFUSIONS TO AVOID

An anomaly score is not a trading signal until its direction, persistence, costs, and false positives are validated.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to Frobenius norm, anomaly detection.

QUICK RECALL

Explain reconstruction error without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Low-rank approximation

INTERMEDIATE

ONE-SENTENCE MEANING

A low-rank approximation retains a small number of dominant directions to summarize a matrix.

MEMORY JOGGER

Replace a complicated surface with its strongest structural modes.

WHY IT MATTERS IN QUANT RESEARCH

It compresses covariance, denoises panels, and reduces model inputs.

FORMULA INTUITION

$X_k = U_k S_k V_k^T$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A 500-asset panel approximated by 10 modes stores common structure while discarding much idiosyncratic noise.

PYTHON / SYSTEM IMPLEMENTATION

Benchmark memory, speed, and out-of-sample error against the full representation.

CONFUSIONS TO AVOID

Low variance does not equal noise; discarded directions can contain relative-value signals.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to Eckart-Young theorem, matrix completion.

QUICK RECALL

Explain low-rank approximation without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Whitening

INTERMEDIATE

ONE-SENTENCE MEANING

Whitening transforms components to have unit sample variance and zero sample covariance.

MEMORY JOGGER

Rotate to PCA axes, then divide each axis by its standard deviation.

WHY IT MATTERS IN QUANT RESEARCH

It can improve optimization for algorithms sensitive to scale, but changes economic magnitude.

FORMULA INTUITION

$Z_{\text{white}} = X_c V \Lambda^{-1/2}$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A component with variance 9 is divided by 3, while one with variance 0.25 is divided by 0.5.

PYTHON / SYSTEM IMPLEMENTATION

Regularize or drop tiny eigenvalues before whitening to avoid explosive amplification.

CONFUSIONS TO AVOID

Whitening magnifies noisy low-variance directions and destroys the variance ranking.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to sphering, decorrelation, standardization.

QUICK RECALL

Explain whitening without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

PCA as denoising

INTERMEDIATE

ONE-SENTENCE MEANING

PCA denoising reconstructs data from selected components and treats the remainder as noise.

MEMORY JOGGER

Keep stable common structure; discard fragile small modes.

WHY IT MATTERS IN QUANT RESEARCH

It can improve covariance estimates and reduce noisy feature redundancy.

FORMULA INTUITION

$X_{\text{denoised}} = X_c V_k V_k^T + \mu$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

Reconstructing returns from stable common modes may reduce microstructure noise before risk estimation.

PYTHON / SYSTEM IMPLEMENTATION

Choose k out of sample and compare downstream metrics, not merely in-sample smoothness.

CONFUSIONS TO AVOID

PCA cannot know whether a low-variance spread is noise or valuable alpha.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to shrinkage, factor model, signal-to-noise.

QUICK RECALL

Explain pca as denoising without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Rolling and expanding PCA

INTERMEDIATE

ONE-SENTENCE MEANING

Rolling PCA refits on a moving window; expanding PCA uses all data observed so far.

MEMORY JOGGER

The basis itself evolves as market structure changes.

WHY IT MATTERS IN QUANT RESEARCH

Time-varying components can reveal regimes but create turnover and comparability problems.

FORMULA INTUITION

$V_t = \text{PCA}(X_{\{t-L+1:t\}})$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A rolling 252-day PCA adapts faster than an expanding fit but has noisier loadings.

PYTHON / SYSTEM IMPLEMENTATION

Fit at scheduled timestamps, lag outputs, and version every basis used by production.

CONFUSIONS TO AVOID

Overlapping windows create smooth-looking series and strong dependence; do not overstate independent evidence.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to window length, online PCA, concept drift.

QUICK RECALL

Explain rolling and expanding pca without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Component instability

INTERMEDIATE

ONE-SENTENCE MEANING

Component instability is variation in estimated directions across samples or windows.

MEMORY JOGGER

Nearby eigenvalues allow directions to rotate even when the subspace is stable.

WHY IT MATTERS IN QUANT RESEARCH

Unstable loadings undermine interpretation, feature continuity, and portfolio turnover control.

FORMULA INTUITION

similarity = $|\mathbf{v}_t^T \mathbf{v}_{t-1}|$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

Two individual components may swap order while their combined two-dimensional subspace barely changes.

PYTHON / SYSTEM IMPLEMENTATION

Monitor eigenvalue gaps, principal angles, sign-aligned cosine similarity, and subspace distance.

CONFUSIONS TO AVOID

Comparing raw signed vectors mistakes harmless sign flips for instability.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to eigenvalue gap, principal angles, Procrustes alignment.

QUICK RECALL

Explain component instability without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Sign ambiguity and alignment

INTERMEDIATE

ONE-SENTENCE MEANING

An eigenvector and its negative represent the same geometric direction.

MEMORY JOGGER

Flipping every loading also flips every score, leaving reconstruction unchanged.

WHY IT MATTERS IN QUANT RESEARCH

Stable reporting and live feature continuity require a deterministic sign convention.

FORMULA INTUITION

v and $-v$ are equivalent. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

Choose the sign so the largest absolute loading is positive, or align to the previous fitted vector.

PYTHON / SYSTEM IMPLEMENTATION

Apply sign alignment after sorting and before storing components; record the rule.

CONFUSIONS TO AVOID

Sign alignment does not solve component swapping or rotation within near-degenerate subspaces.

WHERE THIS FITS

Research path: clean data $\rightarrow$ fit transformation $\rightarrow$ validate out of sample $\rightarrow$ store artifacts $\rightarrow$ monitor drift. This concept connects directly to label switching, basis alignment.

QUICK RECALL

Explain sign ambiguity and alignment without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Outliers and robust PCA

INTERMEDIATE

ONE-SENTENCE MEANING

Outliers can dominate covariance and rotate principal directions toward rare extreme observations.

MEMORY JOGGER

Variance rewards distance, including bad ticks and one-off shocks.

WHY IT MATTERS IN QUANT RESEARCH

Financial data contains jumps, stale prices, and feed errors that can distort PCA.

FORMULA INTUITION

classical PCA minimizes squared reconstruction error. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

One erroneous 50-sigma return can become the main direction in a small sample.

PYTHON / SYSTEM IMPLEMENTATION

Validate data first; compare winsorized, robust-scaled, and robust-covariance variants with an audit trail.

CONFUSIONS TO AVOID

Removing genuine crises as outliers can erase the risk structure the model must understand.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to robust covariance, leverage point, winsorization.

QUICK RECALL

Explain outliers and robust pca without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Missing data and asynchronous markets

INTERMEDIATE

ONE-SENTENCE MEANING

Missing and nonsynchronous observations can corrupt covariance and make component structure inconsistent.

MEMORY JOGGER

PCA needs a coherent rectangular panel or a model designed for incomplete data.

WHY IT MATTERS IN QUANT RESEARCH

Global universes, listings, halts, and stale quotes create structured missingness.

FORMULA INTUITION

pairwise covariance may not be PSD. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

Filling a closed market's return with zero can manufacture correlation patterns around timezone boundaries.

PYTHON / SYSTEM IMPLEMENTATION

Define calendars, stale-price rules, minimum coverage, and train-only imputation; test PSD afterward.

CONFUSIONS TO AVOID

Convenient imputation can add false certainty and leak future values.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to EM-PCA, matrix completion, calendar alignment.

QUICK RECALL

Explain missing data and asynchronous markets without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

PCA and factor models

INTERMEDIATE

ONE-SENTENCE MEANING

PCA yields statistical factors; economic factor models specify or estimate interpretable drivers.

MEMORY JOGGER

One discovers variance directions, the other encodes an economic story.

WHY IT MATTERS IN QUANT RESEARCH

PCA can initialize latent risk factors or benchmark named factor models.

FORMULA INTUITION

$X \approx F B^T + E$. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A first equity component may resemble the market, but the resemblance must be measured rather than assumed.

PYTHON / SYSTEM IMPLEMENTATION

Regress component scores on known factors and compare exposures, residual risk, and stability.

CONFUSIONS TO AVOID

A statistical component is not automatically investable, causal, or stable.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to APT, latent factors, risk model.

QUICK RECALL

Explain pca and factor models without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Leakage-safe research pipeline

INTERMEDIATE

ONE-SENTENCE MEANING

A leakage-safe pipeline fits every data-dependent transform using only information available at the decision time.

MEMORY JOGGER

Preprocessing is part of the model and belongs inside validation folds.

WHY IT MATTERS IN QUANT RESEARCH

Centering, scaling, imputation, PCA, and component selection can all leak future structure.

FORMULA INTUITION

`fit(train); transform(train, validation, test)`. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

For each walk-forward split, refit PCA on the training window and transform the next period without peeking.

PYTHON / SYSTEM IMPLEMENTATION

Use immutable pipeline objects, cutoff timestamps, lineage metadata, and tests that perturb future data.

CONFUSIONS TO AVOID

Fitting PCA once on the full history contaminates every earlier backtest observation.

WHERE THIS FITS

Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to walk-forward validation, purging, embargo.

QUICK RECALL

Explain leakage-safe research pipeline without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

Production monitoring

INTERMEDIATE

ONE-SENTENCE MEANING

Production monitoring checks whether inputs, components, and downstream behavior remain consistent with the fitted system.

MEMORY JOGGER

Monitor the geometry, not just the final PnL.

WHY IT MATTERS IN QUANT RESEARCH

Drift can appear as scale changes, loading rotation, score shifts, or rising reconstruction error.

FORMULA INTUITION

track PSI, score moments, subspace angles, residual error. Read the expression by checking shapes, units, and the direction of the mapping.

SIMPLE MARKET EXAMPLE

A stable forecast with sharply rotating loadings may hide compensating model behavior and growing fragility.

PYTHON / SYSTEM IMPLEMENTATION

Set alert tiers, retain reference windows, and connect each alert to a documented response playbook.

CONFUSIONS TO AVOID

Frequent automatic refits can chase noise; monitoring and retraining are separate decisions.

WHERE THIS FITS

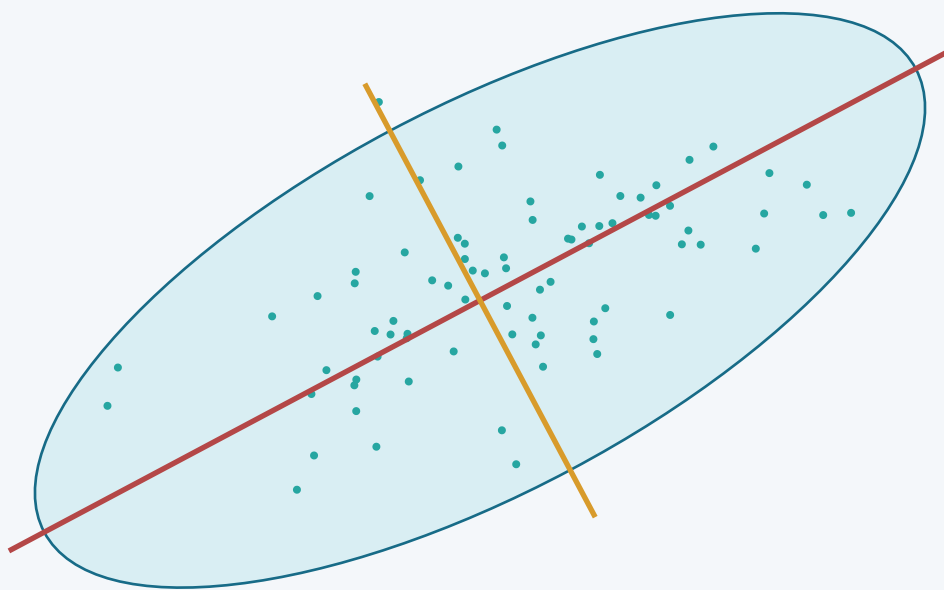
Research path: clean data -> fit transformation -> validate out of sample -> store artifacts -> monitor drift. This concept connects directly to data drift, concept drift, model registry.

QUICK RECALL

Explain production monitoring without notation. Then state one failure mode, one diagnostic, and what must be frozen before validation.

READING COVARIANCE GEOMETRY

The cloud's longest axis is the first eigenvector. Axis lengths are proportional to square roots of eigenvalues. PCA rotates coordinates to these axes.



PC1: maximum variance direction

PC2: orthogonal residual direction

INTERPRETATION CHECKLIST

1. Is the panel centered? 2. Are scales comparable? 3. Does one axis dominate because of economics or units? 4. Are small axes stable across windows? 5. Does the retained subspace help the downstream task?

RECALL

If λ_1 is nine times λ_2 , the standard-deviation axis ratio is 3:1, not 9:1.

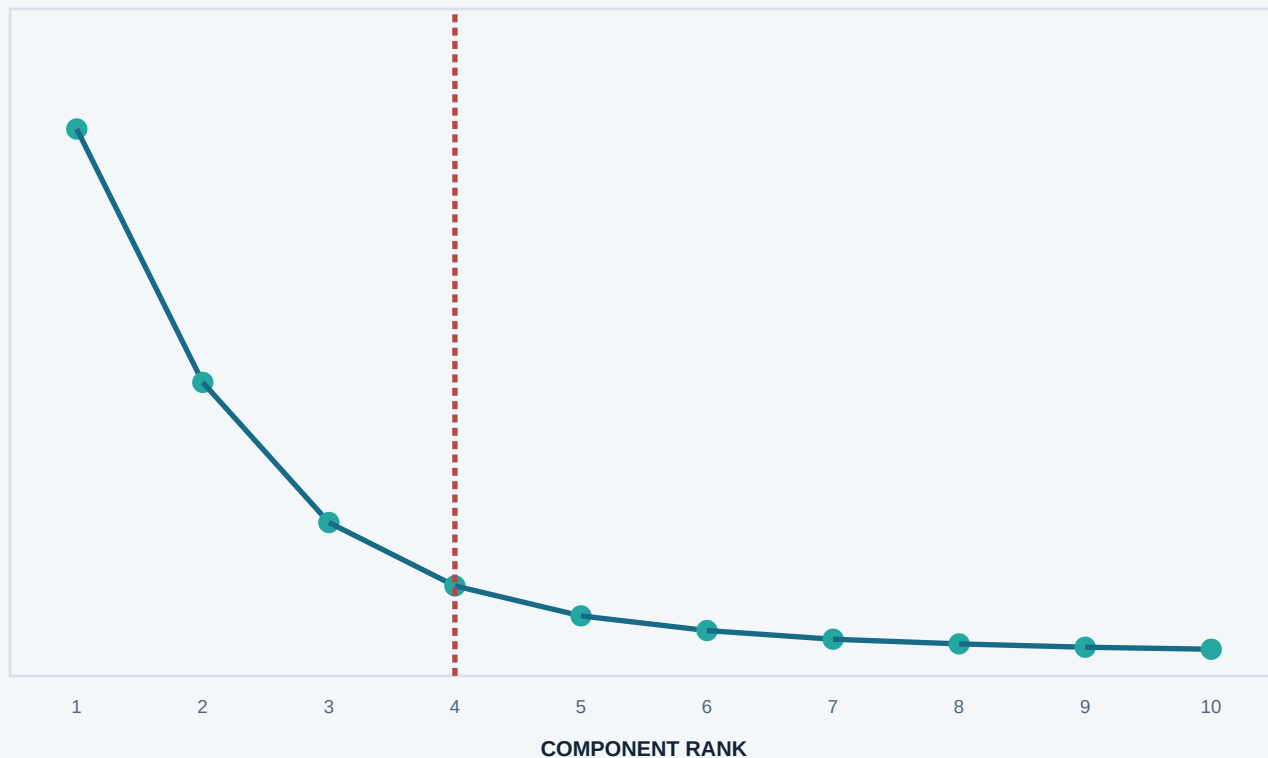
Scree plot and component-count decision

INTERMEDIATE

WHAT THE CHART ANSWERS

How quickly does incremental variance decay, where is the elbow, and whether a compact representation is plausible? It does not determine predictive usefulness.

EIGENVALUE



DECISION OVERLAY

Candidate elbow: around components 3-4. Validate candidate k values with walk-forward reconstruction, downstream loss, loading stability, turnover, and operational cost. Prefer the smallest k that meets the task-specific acceptance criteria.

FAILURE MODES

A dominant unit can create a false elbow. Smooth spectra make visual selection arbitrary. Sampling noise shifts small eigenvalues. Choosing k after seeing test performance is model selection leakage.

SYSTEM FIT

Store the full spectrum, selected k , rule, training cutoff, scaling convention, and validation evidence in the experiment registry. The chart is evidence, not the decision rule by itself.

Regime stability heatmap

INTERMEDIATE

FEATURE STABILITY MAP

Rows are regimes; columns are leading components. Each cell shows explained-variance share. Large changes indicate that the covariance geometry and effective dimension are regime dependent.

Quiet	72%	18%	8%	2%
Trend	54%	25%	14%	7%
Shock	38%	27%	20%	15%
Recovery	61%	21%	11%	7%
	PC1	PC2	PC3	PC4

READ THE PATTERN

Shock periods distribute variance across more directions: PC1 falls from 72% to 38%, while PC3-PC4 rise. A fixed low-rank model may under-reconstruct crises precisely when risk coverage matters most.

VALIDATION ACTIONS

Compare subspace angles, reconstruction error, and downstream loss by regime. Stress component count and estimation window. Set monitoring thresholds using historical variation, not arbitrary percentages.

CAUTION

Regime labels are analytical summaries, not objective truth. If labels use future returns or full-sample clustering, they cannot be used as real-time conditioning variables without a separate online classifier.

Formula review and worked calculation

INTERMEDIATE

CORE IDENTITIES

Center: $X_c = X - \mu$. Covariance: $\Sigma = X_c^T X_c / (T-1)$. Eigenpair: $\Sigma v_i = \lambda_i v_i$. Score: $Z = X_c V_k$.
 Reconstruction: $\hat{X} = Z V_k^T + \mu$.

WORKED VARIANCE CALCULATION

Given eigenvalues [6, 3, 1], total variance is 10. EVR = [60%, 30%, 10%]. Keeping $k=2$ retains 90%. The omitted component contributes 10% of total sample variance, but its predictive value remains untested.

SVD CONNECTION

If $X_c = U S V^T$, then covariance eigenvalues are $s_i^2 / (T-1)$, feature directions are columns of V , and scores are $X_c V = U S$. This avoids explicitly forming a possibly ill-conditioned covariance matrix.

WHITENING CAUTION

Whitened scores divide by $\sqrt{\lambda_i}$. A component with $\lambda=0.0001$ is multiplied by 100, strongly amplifying estimation noise. Drop or regularize weak directions before whitening.

PORTFOLIO CONNECTION

For unit vector v_i , variance is $v_i^T \Sigma v_i = \lambda_i$. Any portfolio w can be decomposed in the eigenbasis, making total variance a weighted sum of eigenmode variances.

RECALL

Derive the dimensions of every expression above. Explain why eigenvalues sum to $\text{trace}(\Sigma)$ and why eigenvector sign does not affect reconstruction.

Failure modes and diagnostics

INTERMEDIATE

SYMPTOM: ONE COMPONENT EXPLAINS ALMOST EVERYTHING

Possible causes: genuine common market mode, one unscaled feature, duplicated columns, or an outlier. Check units, marginal variances, correlations, and robust alternatives.

SYMPTOM: COMPONENTS FLIP OR SWAP

Sign flips are harmless. Swaps often occur when eigenvalues are close. Align signs, compare subspaces with principal angles, and avoid overinterpreting individual vectors in near-degenerate groups.

SYMPTOM: EXCELLENT IN-SAMPLE COMPRESSION, WEAK LIVE MODEL

PCA optimizes reconstruction variance, not target loss. Check leakage, time drift, low-variance signal removal, and whether k was tuned on the test set.

SYMPTOM: UNSTABLE INVERSE COVARIANCE

Inspect small eigenvalues and condition number. Increase sample support, use shrinkage or factor structure, and verify PSD. Do not treat a pseudoinverse as a universal repair.

SYMPTOM: CRISIS RECONSTRUCTION ERROR SPIKES

The retained basis may represent quiet regimes poorly. Evaluate regime-conditioned error, stress larger k , and ensure denoising does not remove tail-risk structure.

TRIAGE ORDER

First validate data and shapes. Then check scaling and time cutoffs. Next inspect spectrum and stability. Only after those pass should you tune the downstream model.

Implementation and model-registry checklist

INTERMEDIATE

DATA CONTRACT

Feature names and order; units; timestamps; calendar; missingness policy; stale-price rule; outlier policy; minimum coverage; universe selection timestamp.

FIT ARTIFACT

Training start and cutoff; means; scales; covariance estimator; component matrix; eigenvalues; selected k; sign convention; software version; random seed where relevant.

VALIDATION EVIDENCE

Walk-forward folds; reconstruction by feature and regime; downstream loss; turnover; subspace stability; sensitivity to windows, scaling, k, and outliers.

RUNTIME CHECKS

Schema hash; finite values; scale drift; score moments; reconstruction error; loading/subspace drift; latency; fallback behavior; alert ownership.

RETRAINING POLICY

Separate monitoring from retraining. Define triggers, minimum data, approval gates, challenger comparison, rollback artifact, and post-deployment observation period.

UNIT TESTS

Known synthetic eigenpairs; orthonormality; trace equality; reconstruction at full rank; invariance to sign flips; no future-data influence; serialization round trip.

Final active recall index

INTERMEDIATE

GEOMETRY

Explain basis, projection, orthogonality, rank, covariance ellipses, eigenvectors, and eigenvalues using one two-asset return cloud.

COMPUTATION

Connect eigendecomposition and SVD. State when to use eigh, why centering matters, and how scores and reconstructions are calculated.

MODEL SELECTION

Give four ways to choose k and explain why explained variance alone is insufficient. Describe leakage-safe nested time-series validation.

ROBUSTNESS

Explain effects of outliers, missing data, asynchronous markets, scaling, short samples, and near-equal eigenvalues.

SYSTEM DESIGN

List every artifact needed to reproduce a live score. Define monitoring for inputs, basis, scores, residuals, and downstream outcomes.

DECISION TEST

Before using PCA, answer: what problem does compression solve, what information may be lost, what baseline will it beat, and what evidence would make you remove it?